Cantera vs NN | Test IC 39 | NRMSE (util.py)

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
2 P-only DeepONet10	**10k**	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	**long**	2.45e-04	0.0148	0.0609	0.0128	3.33e-03	0.0267	2.87e-04	0.0461	0.0488	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	**long**	6.34e-04	0.0365	0.225	0.0833	3.70e-03	0.0233	1.55e-03	0.0378	0.0397	0	Complete
4 T/P DeepONet10	**10k**	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	**long**	0.025	0.0416	0.369	0.0843	0.0523	0.77	0.034	0.205	0.226	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.339	0.38	11.5	0.435	0.582	9.41	0.42	0.91	0.968	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	**long**	4.36e-04	0.0177	0.0281	0.0177	1.90e-03	0.0109	8.75e-04	0.0171	0.0157	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	3.77e-04	0.0501	0.0923	0.119	8.84e-03	0.106	1.76e-03	0.0227	0.0787	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.